

The RF Environment, Explained

The simulated RF world we built for SIH26055 — how it was made from real recordings, how it generates new scenarios, and how we validated it. Companion to *NARADA: The Scanning Agent, Explained*.

1. Why a simulated environment

A receiver can listen to only one narrow band at a time, and must decide many times a second which band to check next. To develop a scheduler for that and compare it fairly against others, we need three things the real world does not give us:

- **Repeatability** — every scheduler must face the identical situation.
- **Ground truth** — to score a scheduler we must know what was transmitting, including in the bands it never checked.
- **Speed** — a 30-second mission runs in a fraction of a second.

The problem statement asks for exactly this: *"a simulated RF environment which has truth information on status of emitters in each band and at each time slot."*

2. The data

The Turing Synthetic Radar Dataset (TSRD), the dataset named in the problem statement: **184 files, 2.2 GB** of recordings of a receiver listening to radar environments. Each scenario comes as a pair — a **scan** recording (receiver sweeping) and a **stare** recording (receiver parked and listening). Each is a list of detected pulses with time, frequency, strength, pulse width and bearing.

Of 92 scenario pairs, **47** are used for development and **45 are kept untouched** for a clean final test. Inside the 47, a **35 / 12** split separates the scenarios the environment builds from and the ones kept for checking, with **zero emitters shared** between them.

3. A generative RF environment

Step 1 — Extract and group. Every recorded pulse is extracted and grouped by the emitter that produced it.

Step 2 — Build a reusable emitter pool. Each emitter's activity becomes a reusable building block, giving **3,443 emitter contributions**.

Step 3 — Sample new scenarios. Emitters are drawn from the pool, with their timing, and dropped together into a fresh 30-second scenario. Every episode is new — the environment can produce combinations of emitters and situations the original dataset never contained.

Step 4 — Fill the grid. Each scenario becomes a grid of **36 bands × 600 time slots**, 50 ms per slot, 30 seconds in total.

The bands

The **36 bands** are the receiver's own, read from each file's metadata and checked against every file opened: centres 500 MHz apart from 250 MHz to 17,750 MHz, each **±500 MHz** wide, so adjacent bands overlap by half. We confirmed this against the data itself: **99.985%** of 4.4 million recorded pulses fall within **±500 MHz** of the active band centre.

What each cell of the grid holds

Quantity	Meaning
$Z[b,t]$	is any emitter transmitting in band b at slot t
$S[b,t]$	the received signal level
$C[b,t]$	how many pulses illuminate the cell
$PW[b,t]$	pulse width of the loudest pulse in the cell
$AOA[b,t]$	angle of arrival of the loudest pulse in the cell

This grid is the ground truth: a complete record of everything, everywhere, at every moment. It is used to **score** a mission, and is never shown to the scheduler. Each emitter's own signal is kept separately too, so a quiet emitter is only credited as found when *its* signal was strong enough — not because a louder neighbour happened to share the band.

4. A realistic receiver

Looking at a band does not reveal the truth — it produces a **noisy measurement**, compared against a detection threshold:

$Y[b,t] = \text{HIT}$ if signal + noise is above the threshold, otherwise NO HIT

Receiver parameter	Value
Detection threshold	-111 dBm
Noise spread	3 dB
Probability of detection (P_d)	0.8506
Probability of false alarm (P_{fa})	0.00135
Sensitivity	-107.16 dBm

So the environment is not noiseless: the receiver occasionally misses a real signal and occasionally raises a false alarm, just as a real receiver does. A scheduler developed here is developed for a realistic detector. The scheduler controls **where** the receiver looks; detection quality belongs to the receiver itself.

5. The rules of the loop

- **Action.** The scheduler picks one of the 36 bands.
- **Dwell.** The receiver listens for that band's natural dwell — 50 ms for 29 bands, 100 ms for the other seven, as in the recordings.
- **Observation.** The scheduler is told HIT or NO HIT, plus what the receiver measured.

- **Repeat.** The clock advances by that dwell; a 30-second mission is **300 to 600 decisions**.

Missions can also run far longer than 30 seconds by stitching fresh scenarios end to end — the basis for continuous online operation.

6. What the scheduler sees, and what it is paid

Seen: only the receiver's own history — per band, how often looking there paid off, how much time it has had, how long since it was last checked, and what was last heard there (strength, pulse width, bearing, pulse count), plus the current band and the mission clock. **Nothing from the truth grid is ever handed over.** This is the problem statement's "*absence of prior reliable intelligence*" requirement, built into the code.

Paid: one fixed reward formula, computed from the same receiver-side quantities:

$$\text{Reward} = \text{exploit what pays} + \text{explore what is overdue} + \text{reward for hits} - \text{cost of hogging airspace}$$

The environment can also take a **per-band priority** from the operator, adding a bonus for discovering emitters on high-threat bands.

7. Validation

Before any scheduler was evaluated, the environment passed **four validation gates**, each with its criteria written down in advance:

Gate	What it checks	Result
1. Out-of-sample prediction	Build truth from stare recordings only, replay the receiver's own scan pattern, and predict what the scan recordings contain — data never used to build it	85.85% correct predictions , MCC 0.685, per-band correlation 0.935
2. Per-band structure	Replay the schedule that produced a recording and compare band-by-band activity	Exact match — 0.35403 vs 0.35403
3. Theory	A controlled periodic emitter against the periodic sweep, versus published closed-form results	Exact match — 0.18333 vs 0.18333
4. Extremes	The emptiest and busiest scenarios, 12 structural checks	12 / 12 pass

Once validated, every physical constant — band layout, 50 ms clock, dwell lengths, noise, threshold — was **frozen** and locked by an automated test, so no result can be improved by quietly changing the world.

8. How schedulers are scored

Three metrics, always reported together:

Metric	Meaning	Better
Interception ratio	fraction of all transmitted pulses the receiver was tuned to	higher
Censored intercept time	average time from an emitter becoming findable to its first detection; an emitter never found is charged the full 30 seconds	lower
Emitter coverage	fraction of findable emitters detected at all	higher

Reporting all three together is what stops a scheduler looking good by camping on one busy band: a camper scores well on interception ratio but badly on intercept time and coverage, and the environment shows it.

Fair evaluation. Every scheduler runs on the identical scenarios with identical noise, and is compared mission by mission rather than by averages, so scenario difficulty cancels out entirely.

9. Glossary

Term	Meaning
Band	One of 36 frequency ranges the receiver can tune to; ± 500 MHz wide, overlapping by half
Slot	50 ms of time. A mission is 600 slots
Dwell	One look at one band — one decision. 50 or 100 ms
Episode / mission	One 30-second run, starting from zero knowledge
Emitter pool	Reusable emitter activity extracted from TSRD, sampled into new scenarios
Ground truth	The complete record of what transmitted where and when; used for scoring only
HIT / NO HIT	The receiver's declaration after one look
Censored	Charging a never-found emitter the full mission length

Built on Gymnasium 1.3 (environment interface), NumPy and h5py 3.16 (TSRD HDF5 access). Figures are from the project's own runs: the four validation gates, the band-geometry measurement and the receiver operating point.